Linear-Extension Sets and Subgroup Rows in S_4: A Finite Witness Separation

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Abstract

We study a finite, self-contained question inside S_4: when a subset of the 24 one-line words is viewed either as a set of linear extensions of a strict poset on [4], or as a subgroup row of S_4, the two structures need not coincide. The paper first fixes the indexing dictionary used for the finite layer. One-line words, local A_3 prefix flags, and 4×4 permutation-matrix supports are canonically bijective, while prefix-support vectors are exactly the cardinality vectors of unordered set-prefix shadows.

For every nonempty $X \subseteq S_4$, the common-precedence relation records the ordered pairs of labels whose relative order is forced in every word of X. We prove that its transitive closure is a strict poset P_X, and that X is an exact poset cone precisely when $X = L(P_X)$. The selected finite data package records bounded package counts and replay checks accompanying this comparison. The mathematical subset count used below is 219 distinct poset-cone subsets from strict labeled posets on [4]; the other selected counts are artifact-row accounting for the packaged rows.

The main witness separation is explicit. A five-word N-poset row is an exact poset cone but not a subgroup row, while the locked D_4 and V_4 rows are subgroup rows whose common-precedence closure is all of S_4, so they are not exact poset cones. Thus the displayed rows witness that exact poset-cone status is not equivalent to subgroup-row status. Left and right coset counts for the subgroup rows are recorded as induced finite group bookkeeping, not as a separate structural equivalence theorem. The paper does not claim a classification of all subsets of S_4, an all-tiler theorem, a geometric realization theorem, or a general S_n theorem.

Keywords: finite symmetric groups; linear extensions; poset cones; subgroup rows; prefix flags; finite certificates.

1 Introduction

The symmetric group S_4 is small enough to permit complete finite replay over its 24 one-line words, but already rich enough to carry different structures on the same ambient labels. A subset of S_4 may be studied as a set of words, as a set of local prefix flags, as a set of permutation-matrix supports, as a linear-extension set of a strict poset, or as a subgroup row. These viewpoints share an index set, but they do not preserve the same mathematical properties.

The purpose of this paper is to prove a controlled finite separation inside that layer. The main comparison is between exact poset-cone status, meaning $X = L(P)$ for a strict poset P on [4], and subgroup-row status, meaning $X \leq S_4$ under the stated composition convention. The prefix-flag and permutation-matrix models provide the finite indexing dictionary; the selected data package provides bounded replay support for the rows and counts used in the argument. Finite certificates are used only for bounded replay claims. They do not replace the proof obligations for the structural lemmas.

External references are used below only for standard terminology around symmetric groups, posets, and linear extensions. The finite claims of this paper are the bounded S_4 statements below.

The main contributions are as follows.

(i) We fix the one-line S_4 convention and prove the canonical bijection between words, local A_3 prefix flags, and 4×4 permutation-matrix supports.

(ii) We identify the A_3 prefix-support vector of a subset $X \subseteq S_4$ with the cardinalities of its unordered set-prefix shadows.

(iii) We prove that the common-precedence closure of a nonempty subset $X \subseteq S_4$ is a strict poset and gives an exact criterion for whether X is a linear-extension set.

(iv) We use the locked D_4, V_4, and N-poset rows to show that exact poset-cone status is not equivalent to subgroup-row status on the displayed finite witnesses.

(v) We record bounded package counts and replay checks accompanying the finite comparison, while separating the 219 distinct mathematical subset count from artifact-row counts.

The scope is deliberately finite. The paper does not classify all 2^{24} subsets of S_4, does not prove an all-tiler theorem, and does not promote a geometric or general S_n theorem. The selected comparison table is a witness table, not a full structural classification of the selected data package.

2 Words, Prefix Flags, and Support

We use standard one-line notation for permutations of the type-A symmetric group; see [BB05] for Coxeter and symmetric-group background. The finite maps below are nevertheless defined explicitly on S_4.

Definition 2.1 (One-line words)**.** Let $[4] = \{1, 2, 3, 4\}$. We write a permutation $\sigma \in S_4$ in one-line notation as

$$\sigma = \sigma_1\sigma_2\sigma_3\sigma_4,$$

where $\{\sigma_1, \sigma_2, \sigma_3, \sigma_4\} = [4]$. Let $\mathcal{W}$ denote this set of 24 one-line words.

Definition 2.2 (Ordered and set-prefix shadows)**.** For $X \subseteq S_4$ and $1 \le r \le 4$, define

$$\mathrm{Pref}_r^{\mathrm{ord}}(X) = \{(\sigma_1, \ldots, \sigma_r) : \sigma \in X\}$$

and

$$\mathrm{Pref}_r^{\mathrm{set}}(X) = \{\{\sigma_1, \ldots, \sigma_r\} : \sigma \in X\}.$$

The ordered prefix shadow remembers the word order; the set-prefix shadow remembers only the underlying support in $[4]$.

Definition 2.3 (Local prefix flags)**.** For $\sigma \in S_4$, set

$$\Phi(\sigma) = \{\sigma_1\} \subset \{\sigma_1, \sigma_2\} \subset \{\sigma_1, \sigma_2, \sigma_3\}.$$

Let $\mathcal{F}$ be the set of all chains $A_1 \subset A_2 \subset A_3 \subset [4]$ with $|A_i| = i$. For a flag $A_1 \subset A_2 \subset A_3$, write its vertex, edge, and face supports as

$$v(A_1, A_2, A_3) = A_1, \qquad e(A_1, A_2, A_3) = A_2, \qquad f(A_1, A_2, A_3) = A_3.$$

The full support $[4]$ is the ambient four-element support and is not counted as a prefix-flag layer.

Definition 2.4 (Prefix-support vector)**.** For $X \subseteq S_4$, define

$$V_{\mathrm{flag}}(X) = \{v(\Phi(\sigma)) : \sigma \in X\}, \quad E_{\mathrm{flag}}(X) = \{e(\Phi(\sigma)) : \sigma \in X\},$$

$$F_{\mathrm{flag}}(X) = \{f(\Phi(\sigma)) : \sigma \in X\},$$

and

$$\mathrm{ps}_{A_3}(X) = \big(|V_{\mathrm{flag}}(X)|, |E_{\mathrm{flag}}(X)|, |F_{\mathrm{flag}}(X)|\big).$$

Lemma 2.5 (Prefix-support dictionary)**.** *For every $X \subseteq S_4$,*

$$\mathrm{ps}_{A_3}(X) = (|\operatorname{Pref}_1^{\mathrm{set}}(X)|, |\operatorname{Pref}_2^{\mathrm{set}}(X)|, |\operatorname{Pref}_3^{\mathrm{set}}(X)|).$$

Proof. Let $\sigma \in X$. By definition,

$$\Phi(\sigma) = \{\sigma_1\} \subset \{\sigma_1, \sigma_2\} \subset \{\sigma_1, \sigma_2, \sigma_3\}.$$

Therefore the vertex support contributed by σ is precisely $\{\sigma_1\}$, the edge support contributed by σ is precisely $\{\sigma_1, \sigma_2\}$, and the face support contributed by σ is precisely $\{\sigma_1, \sigma_2, \sigma_3\}$. Taking the union of these supports over all $\sigma \in X$ gives

$$V_{\mathrm{flag}}(X) = \operatorname{Pref}_1^{\mathrm{set}}(X), \qquad E_{\mathrm{flag}}(X) = \operatorname{Pref}_2^{\mathrm{set}}(X), \qquad F_{\mathrm{flag}}(X) = \operatorname{Pref}_3^{\mathrm{set}}(X).$$

Taking cardinalities gives the stated vector identity. □

Remark 2.6 (Ordered prefixes versus supports). The lemma uses set-prefix shadows, not ordered prefix shadows. For instance, the words `1234` and `2134` have two different ordered length-two prefixes, $(1,2)$ and $(2,1)$, but they determine the same two-element support $\{1,2\}$. The prefix-support vector records supports.

Definition 2.7 (Permutation-matrix support)**.** For $\sigma \in S_4$, define

$$\Delta(\sigma) = \{(i, \sigma_i) : 1 \leq i \leq 4\} \subset [4] \times [4].$$

Let $\mathcal{D}$ be the set of all four-element subsets of $[4] \times [4]$ with exactly one point in each row and exactly one point in each column. Thus an element of $\mathcal{D}$ is the support of a 4×4 permutation matrix.

Theorem 2.8 (The finite S_4 indexing bridge)**.** *The finite sets $\mathcal{W}$, $\mathcal{F}$, and $\mathcal{D}$ are canonically bijective.*

Proof. We first prove that $\Phi : \mathcal{W} \to \mathcal{F}$ is a bijection. For $\sigma \in \mathcal{W}$, the sets

$$\{\sigma_1\} \subset \{\sigma_1, \sigma_2\} \subset \{\sigma_1, \sigma_2, \sigma_3\}$$

have cardinalities $1, 2, 3$, so $\Phi(\sigma) \in \mathcal{F}$.

Define an inverse map $\Psi : \mathcal{F} \to \mathcal{W}$ as follows. Given $A_1 \subset A_2 \subset A_3$ in $\mathcal{F}$, let a_1 be the unique element of A_1, let a_2 be the unique element of $A_2 \setminus A_1$, let a_3 be the unique element of $A_3 \setminus A_2$, and let a_4 be the unique element of $[4] \setminus A_3$. Then

$$\Psi(A_1, A_2, A_3) = a_1 a_2 a_3 a_4.$$

The four labels a_1, a_2, a_3, a_4 are distinct and exhaust $[4]$, so this is a word in $\mathcal{W}$. If A_i came from $\Phi(\sigma)$, then the successive differences are $\{\sigma_1\}$, $\{\sigma_2\}$, $\{\sigma_3\}$, and $\{\sigma_4\}$; hence $\Psi(\Phi(\sigma)) = \sigma$. Conversely, for any flag (A_1, A_2, A_3), the construction gives

$$A_1 = \{a_1\}, \qquad A_2 = \{a_1, a_2\}, \qquad A_3 = \{a_1, a_2, a_3\},$$

so $\Phi(\Psi(A_1, A_2, A_3)) = (A_1, A_2, A_3)$. Thus Φ is a bijection.

We next prove that $\Delta : \mathcal{W} \to \mathcal{D}$ is a bijection. For $\sigma \in \mathcal{W}$, the set $\Delta(\sigma)$ contains exactly one cell in row i, namely (i, σ_i). Since σ is a permutation, the four column values $\sigma_1, \sigma_2, \sigma_3, \sigma_4$ are distinct and exhaust $[4]$; hence every column is used exactly once.

Define $\Omega : \mathcal{D} \to \mathcal{W}$ by reading the unique selected column in each row. If $C \in \mathcal{D}$, let (i, c_i) be the unique cell of C in row i. The condition that each column occurs once says that $c_1 c_2 c_3 c_4$ is a one-line word in S_4, and we set $\Omega(C) = c_1 c_2 c_3 c_4$. Then $\Omega(\Delta(\sigma)) = \sigma$ for every $\sigma \in \mathcal{W}$, and $\Delta(\Omega(C)) = C$ for every $C \in \mathcal{D}$. Thus Δ is a bijection.

Composing the two bijections through $\mathcal{W}$ gives canonical bijections among the three finite models. □

Remark 2.9 (Finite replay). The proof above is structural and does not rely on enumeration. The finite replay table checks the same maps on all 24 words and is used as regression evidence for the computational package.

Remark 2.10 (Indexing boundary). Theorem 2.8 identifies finite labels. It does not assert that every structural property placed on subsets of S_4 is carried over by every later interpretation of those labels.

3 Selected Data Package and Poset-Cone Counts

The finite artifact shipped with the repository uses the historical name "selected atlas." In this paper it is used more narrowly: it is a selected data package for the rows, counts, and witnesses used below, not a classification of all subsets of S_4. Only the 219 distinct poset-cone count is used as a mathematical subset count; the values 226 and 221 are artifact-row accounting values for the selected package boundary.

Definition 3.1 (Selected data row)**.** A selected data row is a finite record attached to a row identifier. An ordinary selected row contains a finite subset $X \subseteq S_4$, written by explicit one-line words, together with declared fields such as row size, prefix data, poset-cone status, subgroup status, and induced coset data when a subgroup row is present. A batch row records a finite family or summary, and is not itself treated as a single subset $X \subseteq S_4$.

Definition 3.2 (Theorem-bearing and diagnostic fields)**.** A field is *theorem-bearing* in this paper only when it is used in a stated result below and has a declared finite replay or proof route. Other recorded fields are diagnostic or deferred fields. Diagnostic fields may help compare rows, but they are not promoted to classification theorems in this paper.

Field group	Typical fields	Role in this paper
Identity and row class	row identifier, name, family, status	locates a finite row or batch record
Underlying subset	size, explicit one-line words	defines $X \subseteq S_4$ for ordinary rows
Prefix data	ordered and set-prefix counts, A_3 prefix-support vector	supports Lemma 2.5 and examples
Poset data	common precedence, closure size, exact poset-cone status	supports the poset-cone counts and Theorem 4.5
Group data	subgroup status, induced left/right coset counts	supports the subgroup count and the locked D_4/V_4 subgroup-row corollaries
Diagnostics and deferred profiles	connectedness, stabilizers, normalizers, deferred profiles	recorded data only, unless a later theorem opens them

Table 1: Compact field-role summary for the selected data package. Not every recorded column is a theorem statement.

Certified Proposition 3.3 (Selected artifact-row count)**.** *Under the declared row schema, the selected data package contains 226 certified artifact rows.*

Proof. The selected data package is a finite artifact with a fixed row schema and a finite summary block. The expected-count certificate and the independent finite checker both record

$$\text{number of selected rows} = 226.$$

The count is a row count inside the selected artifact. It includes ordinary subset rows and two batch summary rows. The row-family accounting used in this paper is displayed in Table 2;

summing the row counts gives

$$2 + 1 + 218 + 1 + 2 + 1 + 1 = 226.$$

This proves the certified row-count statement within the declared selected-data boundary. $\square$

In this count, a strict labeled poset means a transitive irreflexive order relation recorded on the label set [4]; equivalently, it is the strict comparability relation associated with a finite partial order on [4].

Certified Proposition 3.4 (Strict labeled posets and distinct poset cones)**.** *The selected data package records 219 strict labeled posets on* [4]*. Their linear-extension sets give 219 distinct poset-cone subsets of* S_4*.*

Proof. The finite poset enumeration certificate records two equal quantities:

$$\#\{\text{strict labeled posets on } [4]\} = 219$$

and

$$\#\{L(P) : P \text{ a strict labeled poset on } [4]\} = 219.$$

The second quantity is the one used as a subset count: it counts distinct subsets of S_4, not artifact rows. The equality is also consistent with the common-precedence recovery principle proved later in Theorem 4.5: the linear-extension set of a finite strict poset determines exactly the forced order relations of that poset, because incomparable pairs can be realized in either relative order by linear extensions. Thus the certified finite enumeration supplies 219 distinct poset-cone subsets in the selected [4] layer. $\square$

Certified Proposition 3.5 (Exact artifact-row count)**.** *The selected data package records 221 exact poset-cone artifact rows among its 226 selected rows. This 221 is an artifact-row count, not the number of distinct poset-cone subsets; the distinct count is 219.*

Proof. The expected-count certificate records

$$\text{exact poset-cone artifact rows} = 221$$

and Certified Proposition 3.4 records 219 distinct poset-cone subsets. These numbers count different objects.

The selected artifact contains a deduplicated poset-cone layer with 219 distinct subsets. It also contains two sanity rows that are themselves exact poset cones: the singleton row $\{\texttt{1234}\}$ and the full row S_4. These two rows duplicate subsets already present in the deduplicated poset-cone layer. Equivalently,

$$219 \text{ distinct poset-cone subsets} + 2 \text{ duplicate exact sanity rows} = 221$$

exact poset-cone artifact rows. Hence 221 is a row count in the selected artifact, while 219 is the distinct mathematical subset count.

The empty row in the edge-case family is included only as an artifact edge case. It is not an exact poset cone under Definition 4.3, because every finite strict poset on [4] has at least one linear extension; the common-precedence criterion of Theorem 4.5 is stated only for nonempty X. $\square$

Certified Proposition 3.6 (Subgroup count)**.** *Under the locked composition convention used in this paper, the selected data package records 30 subgroups of* S_4*.*

Proof. The finite subgroup enumeration in the selected data package records 30 subgroups of S_4, with subgroup-size distribution

$$1^1,\ 2^9,\ 3^4,\ 4^7,\ 6^4,\ 8^3,\ 12^1,\ 24^1.$$

The weighted sum of this distribution is not the relevant check; rather, the exponents count how many subgroups occur at each order, and their sum is

$$1 + 9 + 4 + 7 + 4 + 3 + 1 + 1 = 30.$$

The same convention is used for the locked D_4 and V_4 subgroup statements in Section 5. □

Family	Rows	Exact rows	Role
edge case	2	1	empty row and singleton sanity row
full group	1	1	full S_4 sanity row
poset cone	218	218	generic deduplicated poset-cone rows
N-poset boundary	1	1	distinguished five-word boundary example
subgroup rows	2	0	locked D_4 and V_4 rows
subgroup batch	1	0	summary of all S_4 subgroups
coset batch	1	0	induced coset summaries for locked subgroup rows
Total	226	221	selected artifact boundary

Table 2: Row-family accounting for the selected data package. Batch rows support finite summaries but are not ordinary subsets $X \subseteq S_4$.

Quantity	Certified value
Selected artifact rows	226
Strict labeled posets on [4]	219
Distinct poset-cone subsets of S_4	219
Exact poset-cone artifact rows	221
Subgroups of S_4 under the locked convention	30

Table 3: Selected package counts recorded for replay context. Only the 219 distinct poset-cone count is a mathematical subset count; 226 and 221 are artifact-row counts.

Remark 3.7 (What the selected data package does not count)*.* The selected data package is not the full power set of S_4. Since $|S_4| = 24$, a full subset classification would contain $2^{24} =$ 16,777,216 subsets. The 226 rows above are a selected finite artifact boundary, not an all-subsets classification.

4 Common Precedence and Poset Cones

We use standard terminology for finite posets and linear extensions; see [Sta12, Tro92]. The common-precedence criterion is proved directly in the finite setting used here.

Definition 4.1 (Common precedence)**.** Let $\emptyset \neq X \subseteq S_4$. For distinct $a, b \in [4]$, write $a\, R_X\, b$ if a precedes b in every one-line word of X. Let P_X be the transitive closure of R_X, considered as a relation on [4]. After Lemma 4.2, define

$$\mathrm{cl}_{\mathrm{poset}}(X) = L(P_X),$$

where $L(P)$ denotes the set of linear extensions of the strict poset P written as one-line words.

Lemma 4.2 (Common-precedence closure is a strict poset)**.** *For every nonempty $X \subseteq S_4$, the relation P_X is a strict poset on* $[4]$.

Proof. By construction, P_X is transitive. It remains to prove that it is irreflexive. Choose any word $\sigma \in X$. Every relation $a\, R_X\, b$ is respected by the total order of the positions in σ: the position of a in σ is smaller than the position of b. Hence every chain in the transitive closure of R_X is also strictly increasing in the position order of σ. A relation $a <_{P_X} a$ would therefore force the position of a in σ to be strictly smaller than itself, impossible. Thus P_X is irreflexive and transitive, hence a strict poset. □

Definition 4.3 (Exact poset cone)**.** A subset $X \subseteq S_4$ is an exact poset cone if there exists a strict poset P on $[4]$ such that $X = L(P)$.

Lemma 4.4 (Finite extension separation)**.** *Let P be a finite strict poset and let a, b be incomparable in P. Then P has a linear extension in which a precedes b, and a linear extension in which b precedes a.*

Proof. It is enough to prove the first assertion; the second is symmetric. Add the relation $a < b$ to the strict relation of P and take the transitive closure. This does not create a cycle: such a cycle would contain a path from b to a already present in P, contradicting the incomparability of a and b. Thus the enlarged relation is again a finite strict poset. Every finite strict poset has a linear extension, for example by repeatedly choosing a minimal remaining element. Any linear extension of the enlarged poset is a linear extension of P in which a precedes b. Repeating the same argument after adding $b < a$ gives a linear extension in the reverse order. □

Theorem 4.5 (Common-precedence exactness criterion)**.** *For every nonempty $X \subseteq S_4$,*

$$X \subseteq L(P_X).$$

Moreover, X is an exact poset cone if and only if

$$X = \mathrm{cl}_{\mathrm{poset}}(X) = L(P_X).$$

Proof. By Lemma 4.2, P_X is a strict poset. If $a\, R_X\, b$, then every word in X places a before b. Hence every word in X respects every relation in R_X. Since each word is linearly ordered, it also respects every relation obtained by transitive closure. Therefore every word of X is a linear extension of P_X, and $X \subseteq L(P_X)$.

Suppose first that X is an exact poset cone, say $X = L(P)$ for a strict poset P on $[4]$. If $a <_P b$, then every linear extension of P places a before b, so $a\, R_X\, b$. Conversely, assume $a\, R_X\, b$. The reverse relation $b <_P a$ is impossible, since then every linear extension of P would place b before a. If a and b were incomparable in P, Lemma 4.4 would give a linear extension of P in which b precedes a, again contradicting $a\, R_X\, b$. Thus $a <_P b$. The common-precedence relation therefore recovers exactly the order relation of P, up to transitive closure, and $P_X = P$. Consequently

$$\mathrm{cl}_{\mathrm{poset}}(X) = L(P_X) = L(P) = X.$$

For the converse, if $X = \mathrm{cl}_{\mathrm{poset}}(X) = L(P_X)$, then X is the set of linear extensions of the strict poset P_X. This is exactly the definition of an exact poset cone. □

5 The Locked D_4 and V_4 Subgroup Rows

Definition 5.1 (Composition and coset convention)**.** In this section a word $p = p_1p_2p_3p_4$ is read as the function $i \mapsto p_i$. Permutations are composed by

$$\mathrm{compose}(p, q) = p \circ q,$$

that is, p after q. For a subgroup $H \leq S_4$, left cosets are written gH and right cosets are written Hg.

Definition 5.2 (Locked rows)**.** The locked rows used in this paper are

$$D_4 = \{\texttt{1234}, \texttt{1324}, \texttt{2143}, \texttt{2413}, \texttt{3142}, \texttt{3412}, \texttt{4231}, \texttt{4321}\}$$

and

$$V_4 = \{\texttt{1234}, \texttt{2143}, \texttt{3412}, \texttt{4321}\}.$$

Corollary 5.3 (The locked D_4 and V_4 rows are not exact poset cones)**.** *The rows D_4 and V_4 in Definition 5.2 are not exact poset cones.*

Proof. For each unordered pair $\{a, b\}$ with $a < b$, the word 1234 places a before b. The following table gives, for each locked row, a word placing b before a.

$\{a, b\}$	D_4 reverse witness	V_4 reverse witness
$\{1, 2\}$	2143	2143
$\{1, 3\}$	3142	3412
$\{1, 4\}$	2413	3412
$\{2, 3\}$	1324	3412
$\{2, 4\}$	3142	3412
$\{3, 4\}$	2143	2143

Thus no ordered pair of distinct labels is a common-precedence relation for either locked row. In both cases P_X is the antichain on $[4]$, and $L(P_X) = S_4$. The locked rows have cardinalities 8 and 4, respectively, so neither row is equal to its common-precedence closure. By Theorem 4.5, neither row is an exact poset cone. □

Corollary 5.4 (The locked rows are subgroup rows with induced coset partitions)**.** *The locked D_4 row is a subgroup row of order 8, so its left and right cosets partition S_4 into three blocks. The locked V_4 row is a subgroup row of order 4, so its left and right cosets partition S_4 into six blocks.*

Proof. For the locked D_4 row, set

$$r = \texttt{2413}, \qquad s = \texttt{1324}.$$

Under the convention of Definition 5.1, direct composition gives

$$r^4 = \texttt{1234}, \qquad s^2 = \texttt{1234}, \qquad srs = r^{-1} = \texttt{3142}.$$

The normal forms generated by r and s are

$$\begin{aligned} &r^0 = \texttt{1234}, \quad r^1 = \texttt{2413}, \quad r^2 = \texttt{4321}, \quad r^3 = \texttt{3142}, \\ &sr^0 = \texttt{1324}, \quad sr^1 = \texttt{3412}, \quad sr^2 = \texttt{4231}, \quad sr^3 = \texttt{2143}. \end{aligned}$$

These are exactly the eight words listed for D_4. Hence the locked D_4 row is the subgroup $\langle r, s\rangle$ of order 8.

For the locked V_4 row, set
$$a = \texttt{2143}, \qquad b = \texttt{3412}.$$
Direct composition gives
$$a^2 = \texttt{1234}, \qquad b^2 = \texttt{1234}, \qquad ab = ba = \texttt{4321}.$$
Therefore
$$V_4 = \{\texttt{1234}, a, b, ab\}$$
is a subgroup of order 4.

Finally, in any finite group, the left cosets of a subgroup partition the group, and the right cosets also partition the group. Since $|S_4| = 24$, the number of left or right cosets is the index $|S_4|/|H|$. Thus the locked D_4 subgroup has $24/8 = 3$ left cosets and 3 right cosets, while the locked V_4 subgroup has $24/4 = 6$ left cosets and 6 right cosets. □

Remark 5.5 (Subgroup separation already visible). Corollaries 5.3 and 5.4 show that subgroup-row status does not imply exact poset-cone status for the selected locked rows. The coset counts are the standard partitions induced by those subgroup rows; they are not used as an independent structural axis.

6 The N-Poset Boundary Example

Example 6.1 (The N-poset row)**.** Let P_N be the poset on $[4]$ with covering relations
$$1 < 3, \qquad 2 < 3, \qquad 2 < 4.$$

Proposition 6.2 (The N-poset witness)**.** *The poset P_N has exactly five linear extensions:*
$$L(P_N) = \{\mathit{1234}, \mathit{1243}, \mathit{2134}, \mathit{2143}, \mathit{2413}\}.$$
Consequently this row is an exact poset cone. It is not a subgroup row of S_4.

Proof. The relations defining P_N say that 1 must precede 3, 2 must precede 3, and 2 must precede 4. Each of the five displayed words satisfies these three requirements.

It remains to see that there are no further linear extensions. At the first step, the minimal elements are 1 and 2. If 1 is chosen first, then 2 must be chosen second, because both 3 and 4 still require 2 to precede them. The remaining labels 3 and 4 may then be ordered freely, giving
$$\texttt{1234}, \qquad \texttt{1243}.$$
If 2 is chosen first, then 1 and 4 are available. The branch beginning with 21 leaves 3 and 4 available in either order, giving
$$\texttt{2134}, \qquad \texttt{2143}.$$
The branch beginning with 24 forces 1 to occur before 3, giving only
$$\texttt{2413}.$$
These five branches exhaust all possible choices of minimal elements, so $L(P_N)$ is exactly the displayed five-word set.

The row is therefore an exact poset cone by definition. It cannot be a subgroup row of S_4, because a subgroup of a finite group has order dividing the group order, and $5 \nmid 24$. □

Remark 6.3 (Role in the comparison). The N-poset row gives an exact poset cone that is not a subgroup row. Together with the locked D_4 and V_4 rows, it supplies the finite witnesses needed in Section 7.

7 Selected Structural Comparison

The preceding sections put two main structural tests on the same ambient set S_4: exact poset-cone status and subgroup-row status. The comparison in this section uses selected witnesses only. It is not a classification of all selected data rows and not a classification of all subsets of S_4.

In Table 4, the induced-cosets column is bookkeeping induced by subgroup status under Definition 5.1. A "yes" in that column means that the row is a subgroup $H \leq S_4$, so its left and right cosets partition S_4. It is not a separate geometric tiling statement and is not used as an independent structural axis.

Witness	Exact	Subgroup	Induced cosets	Role
$\{\texttt{1234}\}$	yes	yes	yes	overlap at the identity row
S_4	yes	yes	yes	overlap at the full group row
N-poset row	yes	no	no	exact poset cone outside subgroup rows
D_4 row	no	yes	yes	subgroup row outside exact poset cones
V_4 row	no	yes	yes	subgroup row outside exact poset cones

Table 4: Selected witnesses used for Theorem 7.1. The induced-cosets column is subgroup bookkeeping. The table compares only the displayed rows and is not a full data-package classification.

Theorem 7.1 (Selected witnesses separate poset-cone and subgroup-row status)**.** *The displayed rows in Table 4 witness that exact poset-cone status and subgroup-row status are not equivalent predicates on subsets of S_4.*

Proof. By Proposition 6.2, the N-poset row is an exact poset cone and has five elements. Since $5 \nmid 24$, it is not a subgroup row of S_4. Thus exact poset-cone status does not imply subgroup-row status.

Conversely, by Corollaries 5.3 and 5.4, the fixed D_4 and V_4 rows are subgroup rows, but neither is an exact poset cone. Hence subgroup-row status does not imply exact poset-cone status.

The two overlap rows in Table 4 record that the fields are not disjoint. The singleton $\{\texttt{1234}\}$ is $L(P)$ for the chain $1 < 2 < 3 < 4$, and $\texttt{1234}$ is the identity subgroup under the group convention. The full row S_4 is $L(P)$ for the antichain on $[4]$, and it is also the full subgroup of itself. These overlap rows are not needed to prove non-equivalence, but they prevent the stronger and false reading that poset-cone and subgroup rows are disjoint classes. $\square$

Remark 7.2 (Boundary of the comparison)*.* Theorem 7.1 separates exact poset-cone status from subgroup-row status. It does not prove a separate pairwise non-equivalence theorem between subgroup-row status and subgroup-coset partition status. In this paper, the induced coset entries for the displayed subgroup rows are the standard left and right coset partitions induced by subgroup status. A full selected-data comparison theorem would require a new row-by-row statement and a separate certificate.

8 Boundaries and Nonclaims

The results above are finite statements about one-line words in S_4, their prefix flags, selected finite data rows, common-precedence poset closures, and subgroup-row witnesses. This section records the boundary of those statements in the same mathematical language, so that the finite conclusions are not read as stronger classification or realization theorems.

Proposition 8.1 (Boundary of the finite result)**.** *The results in this paper do not assert or supply the missing hypotheses for any of the following stronger statements:*

(a) a classification of all 2^{24} subsets of S_4;

(b) a classification of all exact-cover, translate-cover, or tiler behavior in S_4;

(c) a geometric realization or geometric transitivity theorem;

(d) a theorem about special Magic-24, square, or mask structures;

(e) a general theorem for S_n with arbitrary n;

(f) a synthesis theorem identifying all structural fields carried by the same S_4 labels;

(g) a statement that the indexing bridge preserves every structural field placed on subsets of S_4;

(h) an independent theorem separating subgroup-row status from subgroup-coset partition status.

Proof. The indexing bridge of Theorem 2.8 is a bijection among three finite label models: one-line words, local prefix flags, and permutation-matrix supports. A bijection of labels does not by itself define geometric incidence, magic-square constraints, tiler data, or a general S_n construction. Those structures require additional objects and additional compatibility conditions that are not part of the hypotheses of the theorem.

The selected-data statements in Section 3 count a selected finite artifact. They distinguish 226 selected artifact rows, 219 distinct poset-cone subsets, and 221 exact poset-cone artifact rows. Since $|S_4| = 24$, the full power set has 2^{24} subsets. No result above enumerates or classifies that full power set, and no result above gives an implication table for every possible subset of S_4.

The common-precedence criterion of Theorem 4.5 detects when a nonempty subset $X \subseteq S_4$ is exactly the set of linear extensions of its common-precedence poset. This is a poset-cone test. It is not an exact-cover classification, a tiler classification, or a geometric realization theorem.

The subgroup statements for D_4 and V_4 in Section 5 are finite group statements under the stated composition convention. They prove that the fixed rows are subgroup rows with the standard induced left and right coset partitions, and also that they are not exact poset cones. They do not turn coset partition bookkeeping into geometric transitivity, nor do they classify all possible subgroup embeddings or all translate covers.

Finally, Theorem 7.1 is a selected-witness proposition. Its witnesses show that exact poset-cone status is not equivalent to subgroup-row status on the displayed rows. The proposition deliberately does not upgrade the selected witness table into a full classification theorem. Therefore none of the stronger statements listed in the proposition follows from the results proved in this paper. □

Remark 8.2 (How to read the certificates). The certificates support finite replay claims inside their declared boundaries: row counts, fixed word lists, finite closure sizes, subgroup checks, induced coset counts, and selected-witness comparisons. They should not be read as hidden proofs of broader structural classification, geometric realization, Magic-24, or general S_n statements.

9 Related work

The background language of this paper is standard finite combinatorics. The type-A and symmetric-group setting is part of the Coxeter-group framework treated by Björner and Brenti [BB05]. The poset and linear-extension terminology follows the classical treatments of Stanley and Trotter [Sta12, Tro92]. Stanley's order-polytope paper [Sta86] is relevant background for the broader setting in which linear extensions index simplex pieces of order-polytopal objects, although no order-polytope theorem is proved here.

Counting and generating linear extensions are well-studied problems; see Brightwell and Winkler [BW91] and Pruesse and Ruskey [PR94]. The finite rows used here are small enough to be checked directly, but these references place the common-precedence criterion of Theorem 4.5 in the standard linear-extension setting rather than as an isolated convention.

There is also nearby literature on quotients, splittings, and sets of permutations. Björner and Wachs study generalized quotients in Coxeter groups [BW88]. Martin and Sagan introduce a notion of transitivity for groups and sets of permutations [MS06]. These topics are adjacent to subgroup rows and permutation sets, but the claims in this paper are narrower: the fixed D_4 and V_4 rows are used as explicit finite witnesses, not as evidence for a general transitivity, quotient, splitting, or tiling theorem.

This paper is also related in method to the author's type-A poset-cone and extension-tiler work [Bab26]. That work studies larger type-A tiling questions. The present paper instead fixes the rank-three S_4 layer and separates local predicates that can be confused if an indexing bridge is treated as a structural theorem.

10 Conclusion

This paper isolates a finite S_4 layer in which one ambient set of 24 words carries several structures that must be kept separate. The indexing bridge between words, prefix flags, and permutation-matrix supports is canonical as an indexing bridge. It does not imply that exact poset-cone status and subgroup-row status are the same invariant.

The common-precedence criterion gives the poset side of the layer: a nonempty subset $X \subseteq S_4$ is an exact poset cone exactly when it equals the linear-extension set of its recovered common-precedence poset. The locked D_4, V_4, and N-poset rows then provide concrete witnesses showing that exact poset-cone status is not equivalent to subgroup-row status.

Thus the contribution is a controlled finite witness separation. The selected data package supplies bounded counts, explicit rows, and replayable certificate checks for the statements used here, but it is not a full atlas of all S_4 subsets. Any extension to larger symmetric groups, geometric realizations, special square or mask structures, translate-cover classifications, or full subset classifications requires additional definitions and proof data beyond the results proved here.

A Replay Boundary and Data Schema

This appendix records the replay boundary for the finite statements used in the paper. All file paths displayed in this appendix are relative to the public repository root. The replay package is evidence for finite enumeration and consistency checks; it is not a substitute for the proofs in the body of the paper.

A.1 Replay Commands

The read-only replay pass for reviewers is:

```
python scripts/replay_all.py --verify
```

The command verifies certificate status for the packaged finite artifacts, checks the SHA-256 manifest `certified/replay_hashes.sha256`, and runs package certification in read-only check mode. It does not rewrite the replay hash manifest, the package manifest, or release certification. It also does not regenerate the selected S_4 data package; the packaged certificates are fixed finite provenance for this repository.

The maintainer refresh command is:

```
python scripts/replay_all.py --update-manifest
```

This command rewrites `certified/replay_hashes.sha256` and refreshes package certification after intentional source, artifact, table, or certificate changes.

A.2 Reviewer Environment and Expected Output

The replay scripts use the Python standard library. The test suite uses `pytest`, as listed in `requirements.txt`. A reviewer can install the optional test dependency with:

```
python -m pip install -r requirements.txt
```

The expected read-only replay summary is a JSON object with fields `"status": "passed"`, `"mode": "verify"`, `"package_certification_mode": "check"`, and a package-checker summary reporting zero failed checks. The expected public test summary is:

```
9 passed
```

A release build should be made only after the last maintainer refresh and read-only replay pass.

A.3 Finite Values Checked by Replay

The replay package checks the following finite values used in the paper.

A.4 Data Field Roles

The selected data package records more data than the paper promotes to theorems. The field roles are separated as follows.

Batch rows require a separate convention. The selected data package includes batch records for subgroup and coset families. Such rows may support finite summary counts or appendix data, but they are not themselves ordinary subsets $X \subseteq S_4$. Poset-cone statements in the paper apply to ordinary subset rows unless a result explicitly says otherwise.

A.5 Hash Policy

The integrity policy is SHA-256 over the exact bytes of the replay driver and each packaged data, certificate, and table artifact listed in Table 5. The hash manifest itself records those digests but is not self-hashed. The manifest format is

```
SHA256_HEX  repository/relative/path
```

with one line per hash-covered artifact. A paper build is considered tied to the replay package only when:

(i) the reviewer command `python scripts/replay_all.py --verify` exits successfully;

(ii) every hash-covered artifact in Table 5, excluding `certified/replay_hashes.sha256` itself, appears in `certified/replay_hashes.sha256`;

(iii) every hash-covered artifact has the digest recorded in `certified/replay_hashes.sha256`;

(iv) every JSON certificate named in Table 5 has status `passed` where that status field is present;

(v) package certification passes in read-only check mode;

Repository path	Supports	Main finite checks
`scripts/replay_all.py`	replay driver	certificate-status check, hash-manifest verification or refresh, package certification; driver file is hash-covered
`results/s4_bridge_table.json`	Theorem 2.8	24 words, 24 flags, 24 permutation-matrix supports
`certified/s4_bridge_table_certificate.json`	bridge regression certificate	inverse checks and boundary checks all passed
`results/s4_selected_atlas.json`	selected data source	row schema, row families, selected finite records
`certified/s4_selected_atlas_expected_counts.yaml`	Certified Propositions 3.3 to 3.6	226 selected rows, 219 distinct poset-cone subsets, 221 exact artifact rows, 30 subgroups
`certified/s4_selected_atlas_row_certificate.json`	selected row checks	row-level finite invariant checks
`certified/s4_selected_atlas_independent_certificate.json`	independent finite replay	independent finite replay checks passed
`certified/s4_d4_v4_rows_certificate.json`	Corollaries 5.3 and 5.4	fixed D_4 and V_4 rows, closure sizes, subgroup checks, induced coset counts
`results/s4_structural_comparison.json`	Theorem 7.1	8 comparison artifact records: 5 proof witnesses, 1 summary, 2 deferred records
`results/s4_structural_comparison.csv`	tabular replay extract	spreadsheet-readable comparison artifact records
`tables/s4_selected_witness_comparison.md`	appendix/table source extract	5 displayed proof-witness rows
`certified/s4_structural_comparison_certificate.json`	selected comparison certificate	34 checks passed, selected-row boundary enforced
`certified/replay_hashes.sha256`	package integrity	manifest for the replay driver and packaged artifacts; the manifest is not self-hashed

Table 5: Replay paths for the finite claims used in the paper. The paths are part of the replay boundary: renaming or replacing a hash-covered artifact requires the maintainer refresh command and a new verification pass.

Quantity	Value	Used in
One-line words in S_4	24	Section 2
Local prefix flags	24	Theorem 2.8
Permutation-matrix supports	24	Theorem 2.8
Selected artifact rows	226	Certified Proposition 3.3
Strict labeled posets on [4]	219	Certified Proposition 3.4
Distinct poset-cone subsets of S_4	219	Certified Proposition 3.4
Exact poset-cone artifact rows	221	Certified Proposition 3.5
Subgroups of S_4 under the locked convention	30	Certified Proposition 3.6
Fixed D_4 row size	8	Section 5
Fixed D_4 common-precedence closure size	24	Corollary 5.3
Fixed D_4 left and right coset counts	3 and 3	Corollary 5.4
Fixed V_4 row size	4	Section 5
Fixed V_4 common-precedence closure size	24	Corollary 5.3
Fixed V_4 left and right coset counts	6 and 6	Corollary 5.4
Comparison artifact records	8	Theorem 7.1
Displayed comparison proof witnesses	5	Theorem 7.1 and Table 4

Table 6: Finite values checked by the replay package. The comparison artifact contains eight records: five displayed proof witnesses, one summary record, and two deferred records. Only the five displayed witnesses support the stated non-equivalence result.

Field role	Typical fields	Allowed use in this paper
Definition-bearing	labels, word notation, group convention, row identifier, family, size, explicit one-line words	define finite objects and conventions
Theorem-bearing	prefix counts, prefix-support vector, exact poset-cone status, common-precedence relations, closure size, subgroup status, induced coset counts, certified summary counts	support the stated lemmas, theorems, propositions, and corollaries after replay
Diagnostic	ordered/set prefix comparison, chamber-graph connectedness, exact-cover flags beyond subgroup wording, stabilizers, normalizers, component sizes	may be displayed as recorded finite data, but not as classification results
Deferred	deferred profiles and annotations for geometric, square, mask, or larger-family questions	not used as evidence for any theorem in this paper

Table 7: Paper-facing data field roles. The replay package may record diagnostic or deferred fields, but the body of the paper uses only the declared theorem-bearing fields.

(vi) the TeX build is run after the last maintainer refresh.

If any data file, certificate, table, or replay script changes, the maintainer refresh command and TeX build must be rerun, followed by the read-only verification command. The paper statements are then read against the refreshed hash manifest, not against an earlier generated package.

B Selected Witness Extracts

This appendix gives the finite witness extract used by Theorem 7.1. It is derived from `results/s4_structural_comparison.json` and the table extract `tables/s4_selected_witness_comparison.md`. The extract is included only to make the displayed witnesses inspectable; it is not a full selected-data table and does not add a classification claim.

Witness	Size	Exact	Subgroup	Induced cosets	Closure	Role
{1234}	1	yes	yes	yes	1	overlap control
S_4	24	yes	yes	yes	24	overlap control
N-poset row	5	yes	no	no	5	exact poset cone outside subgroup rows
D_4 row	8	no	yes	yes	24	subgroup row outside exact poset cones
V_4 row	4	no	yes	yes	24	subgroup row outside exact poset cones

Table 8: Selected witness status extract. "Induced cosets" means the induced left and right subgroup-coset partition under the convention of Definition 5.1. The table compares only these five displayed witnesses.

Witness	One-line words	Common-precedence relations	ps_{A_3}
{1234}	1234	all six relations of the chain $1 < 2 < 3 < 4$	$(1, 1, 1)$
S_4	all 24 one-line words	$\varnothing$	$(4, 6, 4)$
N-poset row	1234, 1243, 2134, 2143, 2413	$1 < 3,\ 2 < 3,\ 2 < 4$	$(2, 2, 2)$
D_4 row	1234, 1324, 2143, 2413, 3142, 3412, 4231, 4321	$\varnothing$	$(4, 4, 4)$
V_4 row	1234, 2143, 3412, 4321	$\varnothing$	$(4, 2, 4)$

Table 9: Explicit witness data used by the selected comparison. The common-precedence column is the finite relation whose linear extensions give the closure size in Table 8.

The overlap rows show that the poset-cone and subgroup-row fields are not disjoint. The N-poset row gives an exact poset cone that is not a subgroup row. The D_4 and V_4 rows give subgroup rows whose common-precedence closure is all of S_4, hence they are not exact poset cones. The induced coset entries for subgroup rows are finite group bookkeeping. These five rows are exactly the witness set used in Theorem 7.1.

The full selected data package and the structural comparison replay file contain additional rows and diagnostic fields. Those additional records are not promoted here: this appendix records only the finite witnesses needed for the stated separation result.

Data and Code Availability

The public source and reproducibility repository for this release is `https://github.com/consciousfractal/Linear-Extension-Sets-and-Subgroup-Rows-in-S4-A-Finite-Witness-Separation`. The immutable release tag is `v1.0.1`; the release commit is the Git commit targeted by that tag in the public repository. The repository contains the LaTeX source, compiled PDF, finite artifacts, certificates, replay scripts, MIT license, citation metadata, and SHA-256 manifests. Each computational statement used in the paper is tied to a finite artifact named in Appendix A. No archival DOI has been assigned for this release.

References

[Bab26] Oleksiy Babanskyy. Type-a poset cones and extension tilers. `https://github.com/consciousfractal/type-a-poset-cones-and-extension-tilers`, 2026. Public manuscript package; cited as related-method context only.

[BB05] Anders Björner and Francesco Brenti. *Combinatorics of Coxeter Groups*, volume 231 of *Graduate Texts in Mathematics*. Springer, 2005.

[BW88] Anders Björner and Michelle L. Wachs. Generalized quotients in coxeter groups. *Transactions of the American Mathematical Society*, 308(1):1–37, 1988.

[BW91] Graham Brightwell and Peter Winkler. Counting linear extensions. *Order*, 8(3):225–242, 1991.

[MS06] William J. Martin and Bruce E. Sagan. A new notion of transitivity for groups and sets of permutations. *Journal of the London Mathematical Society*, 73(1):1–13, 2006.

[PR94] Gara Pruesse and Frank Ruskey. Generating linear extensions fast. *SIAM Journal on Computing*, 23(2):373–386, 1994.

[Sta86] Richard P. Stanley. Two poset polytopes. *Discrete & Computational Geometry*, 1(1):9–23, 1986.

[Sta12] Richard P. Stanley. *Enumerative Combinatorics, Volume 1*, volume 49 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, second edition, 2012.

[Tro92] William T. Trotter. *Combinatorics and Partially Ordered Sets: Dimension Theory*. Johns Hopkins University Press, 1992.